



SSD, the world's first quad-port SSD, capable of interfacing with up to four systems on chips (SoCs) to centralise storage for software-defined vehicles. It



features single-root input/output virtualisation (SR-IOV), a PCIe generation 4 interface, and an automotive design. The SSD

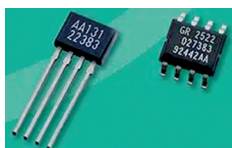
offers speeds for consumer vehicles, with random read and write speeds above 600,000 and 100,000 input/output operations per second (IOPS) respectively, for a 4-kilobyte transfer. This performance allows the drive to manage data streams from multiple SoCs simultaneously, making it suitable for vehicles handling advanced driver-assistance systems (ADAS), in-vehicle infotainment (IVI), and AI-enabled cabin experiences.

Micron Technology

<https://www.micron.com>

Fully integrated inductive switch

Melexis has launched the Induxis switch, the MLX92442. The contactless, magnet-free, and strayfield immune solution directly detects



conductive targets. It supports module designs with fewer components, improv-

ing safety and electrification for high voltage interlocks, charging flaps, seatbelts, hood/trunk, gear tooth sensing, and brake-by-wire applications. The technology directly senses the safety target, such as a seat belt buckle or door latch, without needing a magnet or button actuator, provided it is conductive. This change in position-sensing technology offers several benefits. Without moving parts or additional components, reliability and safety are improved, as the system is not dependent on parts that could become damaged or displaced. Additionally, the solution reduces the bill of materials (BOM) and is more reliable than mechanical solutions, which can

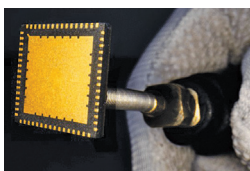
wear out over time, and is immune to magnetic interference.

Melexis

<https://www.melexis.com>

India's first RISC-V MCU chip

Mindgrove Technologies is set to launch India's first 700MHz high-performance microprocessor, Secure IoT, for connected devices that require enhanced chip-level security. Pow-



ered by the Shakti C-class processor developed at IIT Madras, Secure IoT's RISC-V

architecture includes a compressed instruction set integrated with the main instruction set, reducing instruction size while maintaining compatibility. The chip combines processing capabilities with power consumption under 200mW, making it suitable for applications requiring extended battery life, such as smart locks, watches, utility meters, thermal printers, fans, and biometric modules. The company claims it offers extensive peripheral support with 96 functional I/O pins and various communication interfaces. Multiplexing is minimised to include only PWMs, while everything else has dedicated I/O pins. The chip also supports various operating systems and IDEs.

Mindgrove Technologies

<https://www.mindgrovetech.in>

High-precision global navigation satellite system IoT module

Telit Cinterion has launched the SE868K5-RTK module, a high-precision GNSS receiver capable of centimetre-level accuracy. Designed for operation near cellular or other radios, the module offers strong immunity, making it suitable for precision applications. Its 11mm × 11mm form factor allows for use in size-constrained scenarios and easy migration within

the scalable xE868 family. It is a multi-frequency and multi constellation positioning receiver module with real-



time kinematics (RTK) capabilities, enhancing positioning accuracy. Using dual frequencies, it improves loca-

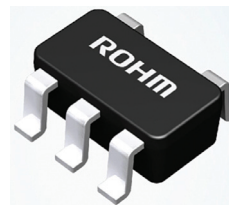
tion precision and reduces multipath effects. The module supports RTK corrections via the RTCM protocol, and raw data output support helps developers manage high-precision applications. Its form factor enables integration across wearables, drones, robots, fleet tracking, and precision agriculture.

Telit Cinterion

<https://www.telit.com>

Ultra-low-power amplifier for sensors

ROHM has developed the LMR1901YG-M, a linear operational amplifier (op-amp) with the lowest current consumption in the industry. It is suitable



for amplifying sensor signals used to detect and measure temperature, flow rate, gas concentration, and other

parameters in battery-powered applications. The op-amp meets the needs for high accuracy and low current consumption by using nano energy technology, resulting in the lowest current consumption on the market. It maintains a typical current consumption of 160nA, about 38% lower than general low-power op-amps. This extends the battery life of devices such as electronic shelf labels and contributes to longer operating times for smartphones and other rechargeable devices. The current consumption remains stable across a temperature range of -40°C to +105°C, ensuring reliable operation in varying conditions, including fire alarms and environmental sensors.

ROHM

<https://www.rohm.com>